

SURIYA PRASATH PARAMESWARAN

DATA SCIENTIST

Leeds, UK · +44 7887 299164 · suriyavictor6@gmail.com · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

UK work authorisation – no sponsorship required (Graduate Route to Feb 2028)

PROFILE

MSc Data Science and Analytics graduate (University of Leeds) who builds machine learning systems that ship and drive a decision, not just score well offline. Work spans fraud detection, customer retention, demand forecasting, and cybersecurity risk, with a focus on calibrated models, capacity-constrained decision frameworks, and deployment via FastAPI, Docker, and MLflow.

EXPERIENCE

Data Science Researcher · University of Leeds (MSc Dissertation)

Sep 2024 — Sep 2025

- Built an end-to-end NLP pipeline classifying CVE severity (LOW/MEDIUM/HIGH/CRITICAL) across ~29,000 NVD records, reaching 0.81 F1 on the CRITICAL class, the highest-stakes tier.
- Benchmarked a fine-tuned SecBERT transformer against a TF-IDF + XGBoost baseline with SMOTE oversampling and SHAP token-level explainability.
- Shipped the model as a Streamlit app with MLflow experiment tracking and GitHub Actions CI/CD.

Tools Python · PyTorch · HuggingFace · SecBERT · XGBoost · SHAP · MLflow · Streamlit

IoT & Data Analytics Intern · Shree Kay Vee Automation

Jan 2023 — Apr 2023

- Analysed IoT sensor data from manufacturing equipment to identify patterns linked to machine performance and failure risk; built dashboards used by the operations team.
- Cleaned and preprocessed sensor data in Python and SQL, improving reporting reliability through structured validation workflows.

PROJECTS

Fraud Triage Engine

2025

- Reframed fraud detection as a capacity-constrained ranking problem on 284,000 transactions; achieved 81.3% fraud recall at 500 daily reviews, a 70% drop in review cost versus naive thresholding.
- Built a calibrated LightGBM + XGBoost ensemble with temporal splitting, cost-sensitive thresholding, and SHAP explainability; a drift monitor triggers automated retraining alerts.
- Shipped as a FastAPI service with Docker and pytest coverage for production reliability.

Tools Python · LightGBM · XGBoost · SHAP · Isotonic Calibration · FastAPI · Docker

Telecom Churn Decision Engine

2025

- Built a profit-optimised retention engine combining T-learner uplift modelling, survival-adjusted CLV, and 0/1 knapsack MILP allocation; beat highest-churn-first targeting by 37.7% mean profit.
- Modelled 8.15x ROI under a fixed £10K retention budget across 2,000 correlated Monte Carlo scenarios.
- Shipped as a FastAPI service with Docker and Monte Carlo stress-testing built in.

Tools Python · LightGBM · scikit-learn · PuLP (CBC MILP) · Monte Carlo · FastAPI · Docker

Rossmann Sales Forecasting

2025

- Ran a forecasting model race (Prophet vs LightGBM vs XGBoost vs CatBoost) across 1,115 stores; XGBoost won with RMSE 1.29M against a 3.09M lag-7 baseline.
- Engineered 18 time-based and store-level features in Polars and added production drift monitoring to flag when forecast quality degrades.

Tools Python · XGBoost · LightGBM · CatBoost · Prophet · Polars · Pydantic v2

EDUCATION

MSc Data Science and Analytics · University of Leeds

Sep 2024 — Sep 2025

Coursework: Applied Machine Learning, Statistical Learning, Natural Language Processing, Data Mining, Time Series Analysis, Cloud Computing, Big Data Analytics.

BE Mechanical Engineering · Easwari Engineering College, Anna University

Jun 2018 — Mar 2022

SKILLS

Programming & Data Python, SQL, pandas, Polars, NumPy, DuckDB, PostgreSQL, MySQL

Machine Learning scikit-learn, LightGBM, XGBoost, CatBoost, SHAP, Optuna, SMOTE, uplift modelling, isotonic calibration, A/B testing, Prophet, ARIMA, MILP (PuLP), Monte Carlo

Deep Learning & NLP PyTorch, HuggingFace Transformers, SecBERT, TF-IDF, transfer learning

MLOps & Deployment FastAPI, Docker, MLflow, GitHub Actions CI/CD, pytest, Streamlit